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Different methods for estimating the mean radiant temperature in an outdoor urban setting

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Abstract:

The mean radiant temperature (T_{mrt}) is one of the most important meteorological parameters governing human energy balance. In this paper, three different methods of obtaining the T_{mrt} in an outdoor urban setting are compared. Method A is based on integral radiation measurements and angular factors, method B is based on measurements with a 38-mm flat grey globe thermometer and in method C makes use of the Rayman 1.2 software is used. Measurements were performed in a large open square in a high latitude city – Göteborg, Sweden – during clear to overcast weather conditions in October 2005 and in July and August 2006.

Results show that the difference between Method A and Method B was generally relatively small. Most of the discrepancy, caused by rapid changes in radiation, temperature and wind speed was smoothed out using 5 min mean values. By systematically and empirically changing the mean convection coefficient, the accuracy of Method B was improved and a new equation expressing the T_{mrt} was obtained. With this new equation the 38 mm flat grey globe thermometer could successfully be used to estimate the T_{mrt} in an outdoor urban setting provided that the wind speed and the air and globe temperatures are measured accurately. The study also shows that the flat grey colour of the globe thermometer slightly underestimates the level of short-wave radiation (i.e. sunshine). Method C works very well during the middle of the day in July, i.e. at high sun elevations. However, the model considerably underestimates the T_{mrt} in the morning and evening in July and during the whole day in October, i.e. at low sun elevations.

In outdoor urban settings where thermal comfort researchers or urban planners and designers require an easy and reliable method of estimating mean radiant temperature, the 38 mm flat grey globe thermometer provides a good and cheap solution. Copyright © 2007 Royal Meteorological Society

KEY WORDS mean radiant temperature; outdoor setting; integral radiation measurements; 38 mm flat grey globe thermometer; Rayman 1.2

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INTRODUCTION

The mean radiant temperature (T_{mrt}), which sums up all short and long wave radiation fluxes (both direct and reflected), to which the human body is exposed is one of the most important meteorological parameters governing human energy balance and the thermal comfort of man. The T_{mrt} is defined as the 'uniform temperature of an imaginary enclosure in which the radiant heat transfer from the human body equals the radiant heat transfer in the actual non-uniform enclosure' (ASHRAE, 2001). There are several methods of measuring and modelling T_{mrt} outdoors. The most accurate, but also the most costly and complex measurement technique, is the performance of integral radiation measurements and the calculation of angular factors (e.g. Höppe, 1992; Spagnolo and de Dear, 2003; Ali-Toudert and Mayer, 2005). A simpler method of measuring the T_{mrt} is to use a globe thermometer (Vernon, 1932; Kuehn *et al.*, 1970; de Dear, 1987; Nikolopoulou *et al.*, 1999). The globe thermometer was first developed for indoor measurements, but has later been applied outdoors (Nikolopoulou *et al.*, 2001). Though simple, mobile and cheap the globe thermometer is seldom used in outdoor comfort studies, mainly due to the lack of outdoor validation. The method has to our knowledge only been tested in controlled test chambers (e.g. de Dear, 1987; Olesen *et al.*, 1989; Nikolopoulou *et al.*, 1999). Over the years, several models have been developed for the calculation of T_{mrt}. The Rayman software (Matzarakis, 2000; Matzarakis *et al.*, 2000) models the T_{mrt}, as well as different thermal indices in the urban structure. The ENVI-met software models the microclimate, including the T_{mrt} in urban structures, and is based on a three-dimensional computational fluid dynamic model and an energy balance model (Bruse, 1999, 2006). Another software developed to model outdoor thermal comfort is the vector-based TownScope model (Teller and Azar, 2001; Azar, 2006). Modelling the T_{mrt} in outdoor spaces however is not evident, particular in complex urban environments and thus all models require simplifications.

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The aim of the paper is to compare different methods of estimating the T_{mrt} in a high latitude outdoor urban setting. Three different methods, two of which are used to measure the T_{mrt} (integral radiation measurements and angular factors, 38 mm grey globe thermometer) and one which is used to model the T_{mrt} (Rayman 1.2 software), are presented and compared. A special focus will be placed on the validation of the 38 mm flat grey globe thermometer as a tool for measuring T_{mrt} in an outdoor setting.

STUDY AREA

Measurements were carried out in a large open square in the city of Göteborg, Sweden (57°42′N, 11°58′E). The city was founded in 1621 and is now the second largest city in Sweden, with nearly 500 000 inhabitants. Figure 1(a) shows an aerial photograph with a view towards the northeast. The open square, marked with an arrow, has a spatial extension of approximately 100 m × 60 m and is located in the city centre. The surrounding built-up area is dense, with buildings 3–4 stories high. A canal runs south of the square, resulting in a rather large fetch in that direction. Figure 1(b) shows a photograph of the square with a view towards the northwest. The surface of the square is flat and covered with light grey granite cobblestones. Buildings on the north and west of the square are made of white plaster. Figure 1(c) shows a photograph of the square towards southeast. The buildings' façades are made of yellow and brown bricks. There is one row of trees at the east end of the square, separating it from the adjacent street.

The Göteborg area has relatively warm winters and cool summers, with mean monthly temperatures of −0.4 °C in February and 16.3 °C in July. The length of the day varies greatly throughout the year due to the high latitude (57°N). In the middle of June, the sun is up for approximately 16 h, but in December the day is only about 6 h long.

(a)

(b)

(c)

Figure 1. (a)Shows an aerial photograph of parts of the Göteborg city core towards northeast. The open square, marked with an arrow. (b)Shows a photo of the square towards northwest and (c)shows a photo of the square towards southeast.

METHODS

Three different methods for the measuring/modelling the T_{mrt} in an outdoor urban setting are compared:

(a) Integral radiation measurements. Calculations of T_{mrt} are based on angular factors for
  (i) a (rotationally symmetric) standing or walking person
  (ii) a sphere
(b) 38 mm flat grey globe thermometer
(c) Rayman 1.2 software (standing or walking person)

Micrometeorological measurements and instrument setup

Micrometeorological measurements were performed during different weather conditions, i.e. from clear to overcast conditions. Measurements were carried out over a total of 5 days between sunrise and sunset: one clear and one overcast day in October 2005, two clear days in July 2006 and one overcast day in August 2006.

Micrometeorological station. A micrometeorological station equipped according to Table I, was used to measure the air temperature, globe temperature, relative humidity, wind speed and wind direction, as well as the three-dimensional short-wave radiation and long-wave radiation flux densities. The measurement height was 1.1 m above the ground, corresponding to the average height of the centre of gravity for adults (Mayer and Höppe, 1987). Wind, relative humidity, air and globe temperature data were sampled every minute and stored in a Campbell CR 10 data logger. Radiation data were stored in a Campbell CR 5000 data logger. The measurements were registered in Central European Time (CET).

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Table I. Measured meteorological variables and instruments.

Variable	Instrument
Air temperature, T_a	Rotronic YA-100
Globe temperature, T_g	AMR Pt100 PK 24
Relative humidity, RH	Rotronic YA-100
Wind speed, Va	R M Young, 8100
Short- and long wave radiation, K, L	Kipp & Zonen, CNR 1

Integral radiation measurements. The most accurate way of determining the outdoor T_{mrt} is to measure the three-dimensional short-wave and long-wave radiation fields along with the angular factors before calculating the T_{mrt} (e.g. Höppe, 1992; Spagnolo and de Dear, 2003; Ali-Toudert and Mayer, 2005). The instrument setup used for the integral radiation measurements is shown in Figure 2. Three net radiometers (Kipp and Zonen, CNR 1), each measuring the four radiation components separately, i.e. the short-wave and long-wave incoming and outgoing radiation fluxes, were mounted on a steel stand in order to measure the three-dimensional radiation field affecting human beings. Short-wave and long-wave radiation fluxes from the four cardinal points, as well as those from the upper and lower hemisphere were measured. The instruments had an offset of approximately 20° from north and were positioned perpendicular to the surrounding building walls.

Globe thermometer measurements. The T_{mrt} can also be measured using a globe thermometer (Vernon, 1932; Bedford and Warner, 1934; Kuehn *et al.*, 1970). Several different models varying in size, thickness and material have been developed over the years. The standard globe thermometer consists of a black-painted copper sphere with a diameter of 150 mm and a thickness of 0.4 mm. It contains a thermometer with its bulb at the centre of the sphere. The globe thermometer used in this study, shown in Figure 3, consists of a hollow acrylic sphere coated in flat grey paint (RAL 7001), with a diameter of 38 mm and a thickness of 1 mm, with a Pt100 sensor at its centre (Humphreys, 1977; de Dear, 1987; Nikolopoulou *et al.*, 1999). The 38 mm flat grey globe thermometer was mounted on the micrometeorological station, next to the air thermometer and within a distance of a 1-m from the wind sensor.

Calculations of the mean radiant temperature

Determination of T_{mrt} by integral radiation measurements. T_{mrt} can be determined if the mean radiant flux density (S_{str}) of the human body is known. In order to calculate S_{str}, the six individual measurements of the short-wave radiation and long-wave radiation fluxes have to be multiplied by the angular factors $F_i (i = 1–6)$ between a person and the surrounding surfaces according to Equation (1) (VDI, 1994):

$$S_{str} = \alpha_k \sum_{i=1}^{6} K_i F_i + \varepsilon_p \sum_{i=1}^{6} L_i F_i \qquad (1)$$

K_i = the short-wave radiation fluxes ($i = 1–6$)
L_i = the long-wave radiation fluxes ($i = 1–6$)

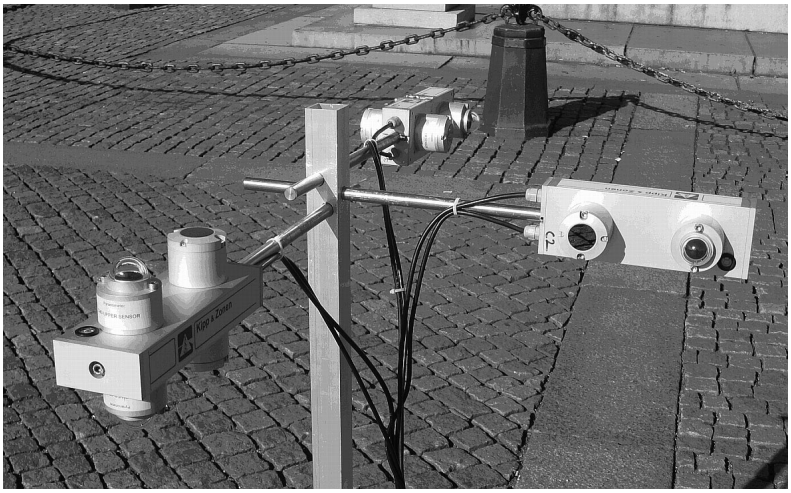

Figure 2. Instrument setup for measuring the three-dimensional short- and long wave radiation field affecting human beings.

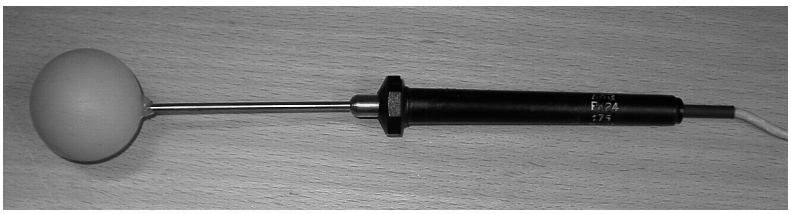

Figure 3. The 38 mm flat grey globe thermometer.

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F_i = the angular factors between a person and the surrounding surfaces ($i = 1$–6)
α_k = the absorption coefficient for short-wave radiation (standard value 0.7)
ε_p = the emissivity of the human body. According to Krichhoff's laws ε_p is equal to the absorption coefficient for long-wave radiation (standard value 0.97)

F_i depends on the position and orientation of the person (Fanger, 1972). The calculation of F_i is complicated for complex urban forms and simplifications are thus necessary. For a (rotationally symmetric) standing or walking person F_i is set to 0.22 for radiation fluxes from the four cardinal points (east, west, north and south) and 0.06 for radiation fluxes from above and below. For a sphere, F_i is 0.167 for all six directions. If S_{str} is known, the T_{mrt} (°C) can be calculated from the Stefan–Boltzmann law:

$$T_{mrt} = \sqrt[4]{\left(S_{str}/\left(\varepsilon_p \sigma\right)\right)} - 273.15 \qquad (2)$$

where:
σ = the Stefan–Boltzmann constant ($5.67 \cdot 10^{-8}$ Wm^{-2} K^{-4})

Determination of T_{mrt} by globe temperature measurements. The theory of the globe thermometer has been thoroughly explained by Kuehn *et al.* (1970). Simply put, the temperature assumed by the globe thermometer at equilibrium results from a balance between the heat gained and lost by radiation and through convection (ASHRAE, 2001). In effect, the globe temperature represents the weighted average of radiant and ambient temperatures. If the globe temperature, air temperature and air velocity are known then the T_{mrt} can be calculated according to Equation (3):

$$T_{mrt} = \left[(T_g + 273.15)^4 + \frac{1.1 \times 10^8 V_a{}^{0.6}}{\varepsilon\ D^{0.4}} \times (T_g - T_a)\right]^{1/4} - 273.15 \qquad (3)$$

T_g = the globe temperature (°C)
V_a = the air velocity (ms^{-1})
T_a = the air temperature (°C)
D = the globe diameter (mm)
ε = the globe emissivity

The empirical derived parameter 1.10×10^8 and the wind exponent ($V_a{}^{0.6}$) together represent the globe's mean convection coefficient (1.10×10^8 $V_a{}^{0.6}$).

Determination of T_{mrt} by the Rayman model. The Rayman 1.2 software (Matzarakis, 2000; Matzarakis *et al.*, 2000) is a tool for the calculation of T_{mrt} and thermal indices such PET, PMV and SET* in urban structures. To calculate T_{mrt}, the programme requires information about the time of day and year, geography (location, altitude and time zone), building geometry (length, width and height), trees (type, height, width of canopy), meteorology (global solar radiation or cloud cover, air temperature and humidity), the albedo of the surrounding surfaces, the Bowen-ratio and the ratio of diffuse and global radiation. The input parameters used in this study were location, time of day and year, 1 min averages of global radiation, air temperature and relative humidity. Default values were used for the albedo (0.3), the Bowen-ratio (1.3) and the ratio of diffuse and global radiation (0.2). Since the input data were site–specific, i.e. measured at the site of interest, no information about building geometry and vegetation were included (Matzarakis, 2004).

RESULTS

Daily three-dimensional short-wave radiation and long-wave radiation pattern and T_{mrt} at a square

A clear summer day. The 26th of July, 2006 was a clear, warm and calm summer day, with a daily mean air temperature of 24.5 °C (maximum 29.1 °C) and a mean wind speed of 1.1 ms^{-1}. From early morning until 8 a.m. the square was shrouded in fog. The sun rose at 3 : 54 a.m. and set at 8 : 42 p.m. The solar elevation reached its maximum of 57.9° at 12:21.

Figure 4(a) shows the six short-wave radiation fluxes K_{east}, K_{west}, K_{south}, K_{north}, $K\downarrow$ and $K\uparrow$, i.e. the short-wave radiation fluxes from the four cardinal points as well as from the upper and lower hemisphere. The incoming short-wave radiation $K\downarrow$, which is controlled by the azimuth and zenith angles of the sun relative to the horizon, reached its maximum of 765 Wm^{-2} at the local solar noon. The reflected short-wave radiation $K\uparrow$, which depends on the amount of incident radiation and the surface albedo, followed the same daily pattern as $K\downarrow$, reaching its maximum of 132 Wm^{-2} at the same time that $K\downarrow$ reached its maximum. K_{east} reached its highest value, 641 Wm^{-2}, in the early morning, when the site was sunlit from east, while K_{west} reached its highest value, 745 Wm^{-2}, in the afternoon, when the site was sunlit from west. K_{south} reached its maximum of 680 Wm^{-2} 11 at a.m., which was about one and a half hours before $K\downarrow$ reached its maximum. K_{north} was low throughout the entire day until 4 p.m., when the site was sunlit. The sun set behind the western buildings at around 6 : 30 p.m., resulting in a sharp decrease in K_{east}, K_{west}, K_{south}, K_{north}, $K\downarrow$ and $K\uparrow$.

Figure 4(b) shows the six long-wave radiation fluxes L_{east}, L_{west}, L_{south}, L_{north}, $L\downarrow$ and $L\uparrow$. In the early morning of the 26th of July 2006 when the square was shrouded in fog, the incoming long-wave radiation, $L\downarrow$ was relatively high (maximum 402 Wm^{-2}) as a result of the reflected infrared radiation back to the surface. $L\downarrow$ later decreased when the fog cleared at around 7 : 30 a.m. In the absence of clouds, $L\downarrow$ is dependent on the bulk atmospheric temperature and emissivity in accordance with the Stefan-Boltzmann Law.

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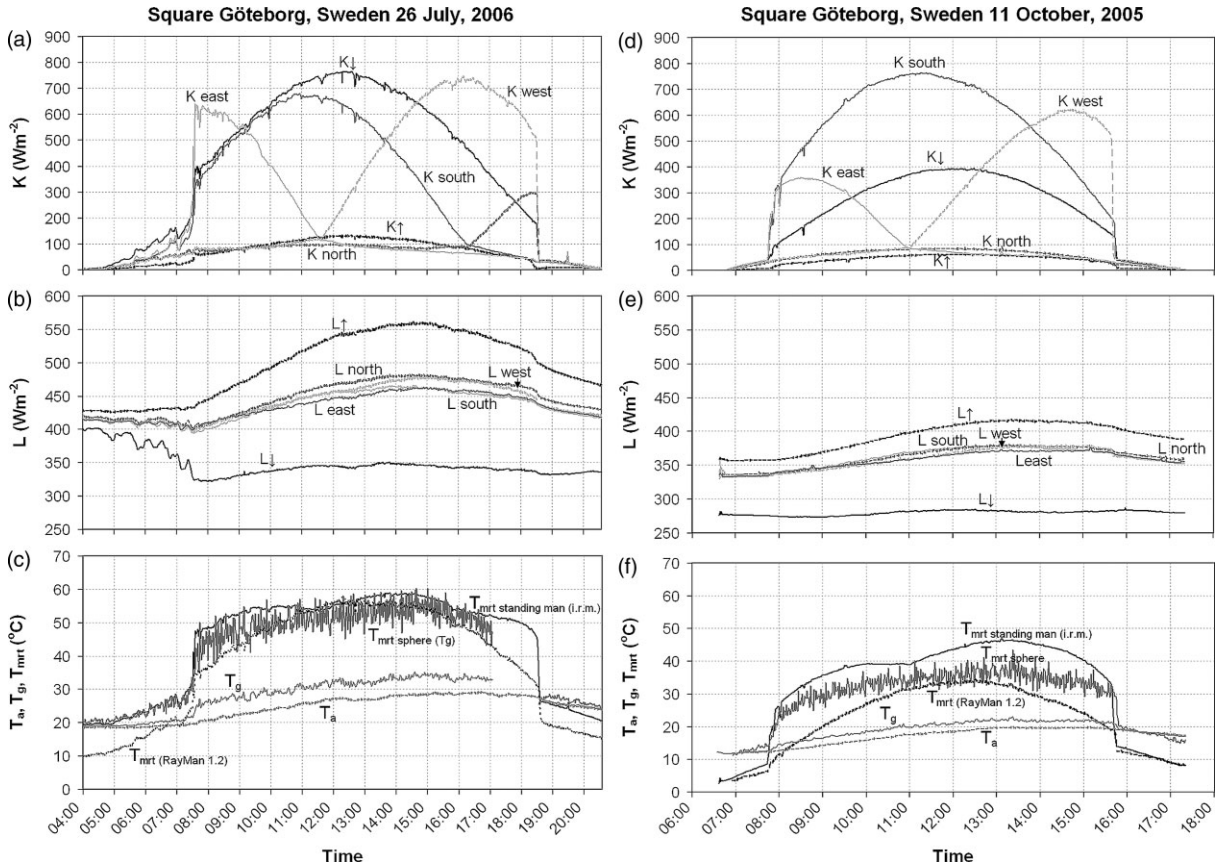

Figure 4. (a)–(c) Three-dimensional short wave and long wave fluxes, air temperature, globe temperature and calculated mean radiant temperature at 1.1 m above ground on a clear summer day (26 July 2006) at a large open square in Göteborg, Sweden (57°N). In the early morning the square was shrouded in fog. (a) Short-wave radiation fluxes, K_{east}, K_{west}, K_{south}, K_{north}, $K\downarrow$ and $K\uparrow$ (b) Long wave radiation fluxes L_{east}, L_{west}, L_{south}, L_{north}, $L\downarrow$ and $L\uparrow$ and (c) T_a, T_g and T_{mrt}, determined by integral radiation measurements and angular factors, $T_{mrt\ standing\ man\ (i.r.m.)}$, 38 mm flat grey globe temperature measurements, $T_{mrt\ (Tg)}$ and Rayman 1.2 software, $T_{mrt\ (Rayman1.2)}$. Figure 4(d–f) Three-dimensional short wave and long wave fluxes, air temperature, globe temperature and calculated mean radiant temperature at 1.1 m above ground on a clear day (11 October 2005) at a large open square in Göteborg, Sweden (57°N). (d) Short-wave radiation fluxes, K_{east}, K_{west}, K_{south}, K_{north}, $K\downarrow$ and $K\uparrow$ (e) Long wave radiation fluxes L_{east}, L_{west}, L_{south}, L_{north}, $L\downarrow$ and $L\uparrow$ and f) T_a, T_g and T_{mrt}, determined by integral radiation measurements and angular factors, $T_{mrt\ standing\ man\ (i.r.m.)}$, 38 mm flat grey globe temperature measurements, $T_{mrt\ (Tg)}$ and Rayman 1.2 software, $T_{mrt\ (Rayman1.2)}$.

Since neither of these properties fluctuates rapidly, $L\downarrow$ is almost constant throughout the day. The outgoing long-wave radiation value, $L\uparrow$, which is governed by surface temperature and emissivity, was both higher and more variable than $L\downarrow$. As the surface temperature increased throughout the day, $L\uparrow$ increased and reached its maximum of 560 Wm^{-2} in the afternoon. L_{east}, L_{west}, L_{south} and L_{north} reached their daily maxima (463 Wm^{-2}, 478 Wm^{-2}, 466 Wm^{-2} and 482 Wm^{-2}) in the early afternoon. As shown, L_{north} was slightly higher than L_{east}, L_{west} and L_{south}. This was because the south facing wall was sunlit during almost the entire day.

Figure 4(c) shows T_{mrt} calculated by Method A ($T_{mrt\ standing\ man\ (i.r.m.)}$), Method B ($T_{mrt\ (Tg)}$) and Method C ($T_{mrt\ (Rayman\ 1.2)}$), along with measured values of T_a and T_g. As shown, the $T_{mrt\ standing\ man\ (i.r.m.)}$ (solid black line) and the $T_{mrt\ (Tg)}$ (solid dark grey line) reached their highest values in the afternoon, between 2 and 3 p.m. These were 58.8 °C and 60.2 °C respectively. The $T_{mrt\ (Rayman\ 1.2)}$ (black dotted line) reached its highest value of 57.5 °C at the local solar noon. As shown, a local $T_{mrt\ standing\ man\ (i.r.m.)}$ minimum was observed around noon. This local minimum is due to the orthogonal instrument setup (i.e. increased mean instrumental error with high angles of incidence Kipp & Zonen, 2002). As shown, the difference between Method A and B is generally relatively small, particularly in the transition from shady to non-shady conditions. The two methods follow the same pattern, although Method B underestimates T_{mrt} and fluctuates rapidly over time. Method C works very well during the middle of the day, however it considerably underestimates T_{mrt} in the morning and afternoon. $T_{mrt\ standing\ man\ (i.r.m.)}$, $T_{mrt\ (Tg)}$ and $T_{mrt\ (Rayman\ 1.2)}$ are about 30, 32 and 30 K higher than T_a (grey dotted line) at the time of their maxima. In the early morning and late evening, $T_{mrt\ standing\ man\ (i.r.m.)}$ and $T_{mrt\ (Tg)}$ were nearly equal to T_a and T_g (solid grey line). Note that T_g and $T_{mrt\ (Tg)}$ are absent between 5 : 05 and 6 : 32 p.m.

A clear autumn day. The 11th of October, 2005 was a relatively warm and calm day for the season, with a daily mean air temperature of 16.9 °C (maximum 19.8 °C) and a mean wind speed of 1.7 ms^{-1}. The skies were clear throughout the day. The sun rose at 6 : 38 a.m. and set at

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5 : 19 p.m. The solar elevation reached its maximum of 30.3° at 12 : 01 p.m.

Figure 4(d) shows the six short-wave radiation fluxes K_{east}, K_{west}, K_{south}, K_{north}, $K\downarrow$ and K from sunrise to sunset on the 11th of October, 2005. $K\downarrow$ and $K\uparrow$ reached their maximum values of 394 Wm^{-2} and 62 Wm^{-2} at the local solar noon. K_{east} reached its highest value, 361 Wm^{-2}, in the morning when the site was sunlit from the east and K_{west} reached its highest value, 622 Wm^{-2}, in the afternoon, when the site was sunlit from the west. On average, K_{south} was twice as large as $K\downarrow$, which was due to the low sun elevation in Göteborg in October. K_{south} reached its maximum of 764 Wm^{-2} around noon, which was about one hour before $K\downarrow$ reached its maximum. At 3 : 40 p.m., the sun set behind the western buildings, resulting in a sharp decrease in K_{east}, K_{west}, K_{south}, K_{north}, $K\downarrow$ and $K\uparrow$.

Figure 4(e) shows the six long-wave radiation fluxes L_{east}, L_{west}, L_{south}, L_{north}, $L\downarrow$ and $L\uparrow$. As shown, the $L\downarrow$ was almost constant and relatively small (273–288 Wm^{-2}) during the day. $L\uparrow$ became larger as the surface temperature increased throughout the day and reached its maximum of 417 Wm^{-2} in the early afternoon. The values of L_{east}, L_{west}, L_{south} and L_{north} were fairly similar (330–380 Wm^{-2}) and they all reached their daily maxima in the afternoon. L_{west}, L_{south} and L_{north} reached their daily maxima around 1 p.m.; however, L_{east} reached its maximum about 2 h later. The L_{west}, L_{south} and L_{north} maxima coincided rather well with the T_a maximum. However, L_{east} reached its maximum (375 Wm^{-2}) when the northern wall was sunlit and K_{west} was at its highest.

Figure 4(f) shows the values of T_{mrt} calculated by Method A ($T_{mrt\ standing\ man\ (i.r.m.)}$), Method B ($T_{mrt\ (Tg)}$) and Method C ($T_{mrt\ (Rayman\ 1.2)}$), along with measured values of T_a and T_g. As shown, $T_{mrt\ standing\ man\ (i.r.m.)}$ and $T_{mrt\ (Tg)}$ reached their highest values between 1 and 2 p.m. These were 46.9 and 43.5 °C respectively. $T_{mrt\ (Rayman\ 1.2)}$ reached its highest value, 34.3 °C, at local solar noon. As shown, the difference between Methods A and B was larger in October in comparison to the values obtained in July (Figure 4(c)). Method C underestimates the T_{mrt} considerably throughout the entire day, except in the early morning and late evening when the measurement site was in shade. $T_{mrt\ standing\ man\ (i.r.m.)}$, $T_{mrt\ (Tg)}$ and $T_{mrt\ (Rayman\ 1.2)}$ are about 27, 24 and 15 K higher than T_a at the time of their maxima.

Summary. The two examples given above represent clear days in the summer and autumn respectively. In October, a large proportion of the short-wave radiation comes from the south, west and east cardinal points. For example, on the 11th of October 2005, K_{south} was, on average, twice as large as $K\downarrow$; this was due to the low sun elevation in Göteborg at this time of the year. The relatively large amount of radiation from the cardinal points results in a relatively high T_{mrt} for a standing man (greater projected area to the side than to the sky and the ground), even when the amount of incoming solar radiation is low.

In general, the difference between Methods A and B was relatively small in July during the entire day. Although Method C worked very well during the middle of the day in July, it underestimated T_{mrt} considerably in the morning and evening. In October, the difference between Methods A and B and C was larger than in July. Both methods (B and C) underestimated T_{mrt} when the measurement site was sunlit; however Method C underestimated T_{mrt} more than Method B.

Validation of the 38 mm grey globe thermometer in an outdoor setting

The influence of weather and response time. To study the influence of weather and response time, the calculated T_{mrt} for a standing person, $T_{mrt\ standing\ man\ (i.r.m.)}$ (Method A) was compared to the T_{mrt} calculated from 38 mm flat grey globe temperature measurements ($T_{mrt\ (Tg.)}$) (Method B). All five days of measurements were included in the analyses. The results presented in Figure 5 show that the difference between the two methods is relatively small. However, there is a large amount of scattering, particularly during semi-cloudy conditions (Figure 5(g–i)). Using 5 min mean values considerably reduces the difference between the two methods and the effect of rapid change in radiation fluxes due to semi-cloudy conditions and wind is nearly completely diminished (Figure 5(b), (e), (h)). Figure 5(c), (f) and (i) also show that the use of 10 min mean values only slightly decreases differences and scattering. Five-minute mean values were therefore used in the subsequent analyses presented in this paper.

The influence of shape. Method A, which is the most accurate method for estimating T_{mrt} outdoors, takes the shape of the body into account. In order to analyse the influence of shape, the difference between the T_{mrt} for a sphere ($T_{mrt\ sphere\ (i.r.m)}$) and a standing man ($T_{mrt\ standing\ man\ (i.r.m)}$) was calculated using Method A. In Figure 6(a), this difference is related to the ratio $K\updownarrow/K_{tot}$ during clear skies and non-shaded conditions. The ratio $K\updownarrow/K_{tot}$ was chosen as a measure of sun elevation. The results presented in Figure 6(a) illustrate that $T_{mrt\ sphere\ (i.r.m)}$ is lower than $T_{mrt\ standing\ man\ (i.r.m)}$ when $K\updownarrow/K_{tot}$ is less than 0.36. When $K\updownarrow/K_{tot}$ is larger than 0.36, $T_{mrt\ sphere\ (i.r.m)}$ is higher than $T_{mrt\ standing\ man\ (i.r.m)}$. This means that the difference between $T_{mrt\ sphere\ (i.r.m)}$ and $T_{mrt\ standing\ man\ (i.r.m)}$ depends on the proportion of vertical radiation and thus the angle of the incident solar radiation.

Figure 6(b) shows the difference between $T_{mrt\ (Tg)}$ (Method B) and $T_{mrt\ standing\ man\ (i.r.m.)}$ (Method A) in relation to the ratio $K\updownarrow/K_{tot}$. The results show that the difference between the two methods may be explained by the differences in shape, to some extent. However, the high amount of scattering indicates the influence of other factors, such as instrumentation and material characteristics.

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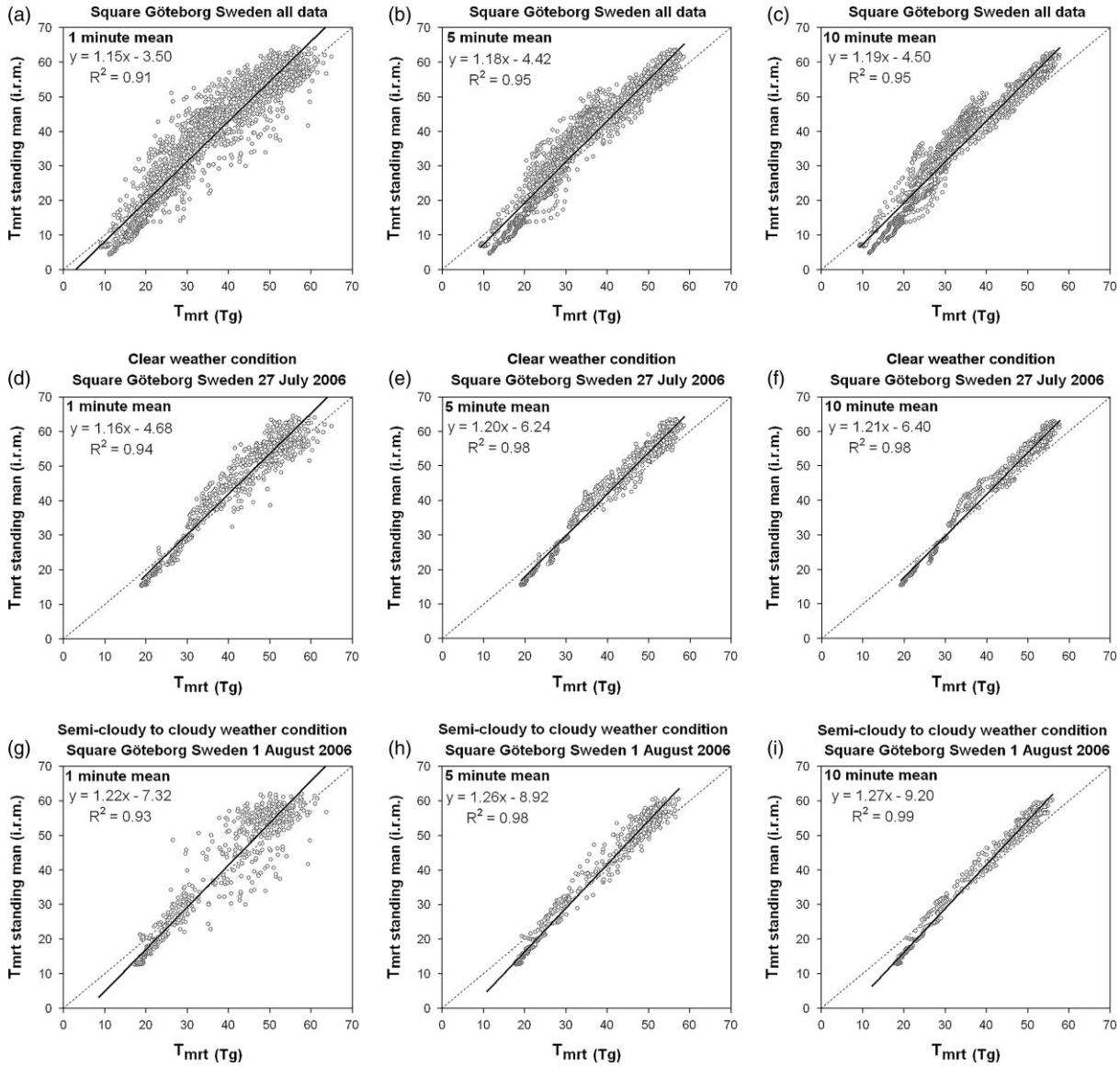

Figure 5. Mean radiant temperature determined from integral radiation measurements and angular factors, $T_{\text{mrt (i.r.m.)}}$ *versus* mean radiant temperature determined from the 38 mm flat grey globe temperature measurements, $T_{\text{mrt (Tg)}}$ during (a)all data 1 min mean (b)all data 5 min mean (c)all data 10 min mean (d)clear weather conditions 1 min mean, (e)clear weather conditions 5 min mean, (f)clear weather conditions 10 min mean, (g)semi-cloudy weather conditions 1 min mean, (h)semi-cloudy weather conditions 5 min mean and (i)semi-cloudy weather conditions 10 min mean.

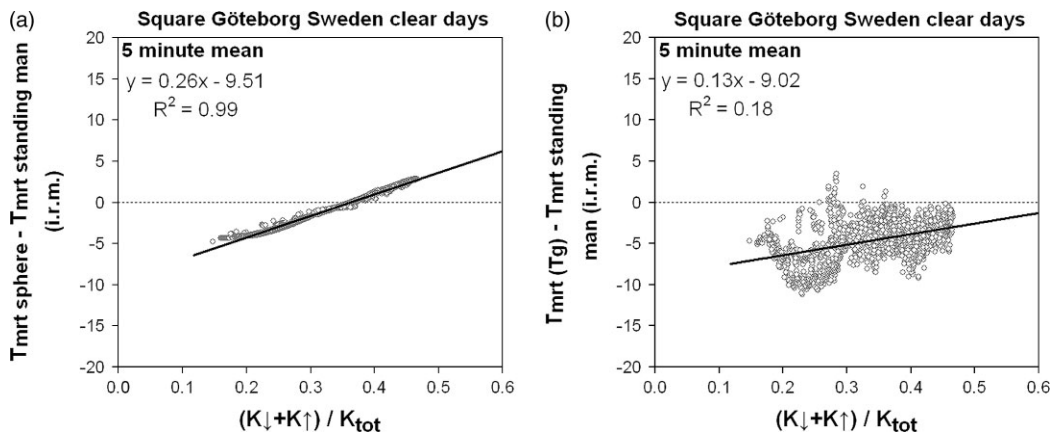

Figure 6. The influence of shape. (a)The difference between the T_{mrt} for a sphere and a standing man, calculated using Method A *versus* the ratio $K\updownarrow/K_{\text{tot}}$ during clear sky and non-shaded conditions. (b)The difference between $T_{\text{mrt (Tg)}}$ calculated using Method B and $T_{\text{mrt standing man (i.r.m.)}}$ calculated using Method A *versus* ratio $K\updownarrow/K_{\text{tot}}$ during clear sky and non-shaded conditions.

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Material characteristics. As shown in Figures 5 and 6(b), the general trend is that Method B underestimates T_{mrt}. From Figure 7, it is evident that Method B generally underestimates T_{mrt} in non-shade conditions but overestimates T_{mrt} in shaded conditions. It is also seen that the scattering is less in shady than in non-shady conditions.

By systematically and empirically changing the mean convection coefficient ($1.10 \times 10^8\ V_a^{0.6}$) in Equation (3), it was possible to analyse the influence of the 38 mm flat grey globe's material characteristics. Figure 8(a) shows the relation between the difference between $T_{mrt\ (Tg)}$ (Method B) and $T_{mrt\ sphere\ (imr)}$ (Method A) and the wind speed on clear days. As shown, Method B underestimates the T_{mrt}. Wind speed also has a small amount of influence. The mean convection coefficient was systematically and empirically adjusted to give zero difference between the two methods as shown in Figure 8(b). Zero difference was obtained at the mean convection coefficient of $1.335 \times 10^8 \cdot V_a^{0.71}$. Inserting the new mean convection coefficient in Equation (3) gives:

$$T_{mrt} = \left[(T_g + 273.15)^4 + \frac{1.335 \times 10^8 V_a^{0.71}}{\varepsilon\ D^{0.4}} \times (T_g - T_a) \right]^{1/4} - 273.15 \qquad (4)$$

The 95% confidence interval for the difference between Methods A and B (Equation 4) is approximately $\pm 3.5\ K$.

A plot of ΔT_{mrt} (Method B-A) against the incoming short-wave radiation ($K\downarrow$) is shown in Figure 9. In Figure 9(a), ΔT_{mrt} is calculated according to Equation (3), while Figure 9(b) shows the results based on the new mean convection coefficient given above (Equation 4). The results show that there is still a systematic difference after adjusting for the shape and material characteristics of the globe, depending on the colour (albedo). As shown, the flat grey colour of the globe causes the influence of short-wave radiation (e.g. sunshine) to be underestimated, i.e. the globe's albedo is too low.

Figure 10 shows the calculated T_{mrt} using the three different methods. Method B is calculated using the new Equation (4), using 5 min mean values. As shown, the difference between Methods A and B is small during the entire day both in July and October, but is slightly greater in October than in July. This is due to the low sun elevation and the difference in shape between the two methods.

A sensitivity test was conducted on Method C (Rayman 1.2) using a stepwise change of the input parameters – the albedo, the Bowen-ratio and the ratio of the diffuse and global radiations. Although the magnitude of T_{mrt} changed, the daily pattern remained the same, i.e. the model still underestimated T_{mrt} in the morning and evening in July and throughout the entire day in October.

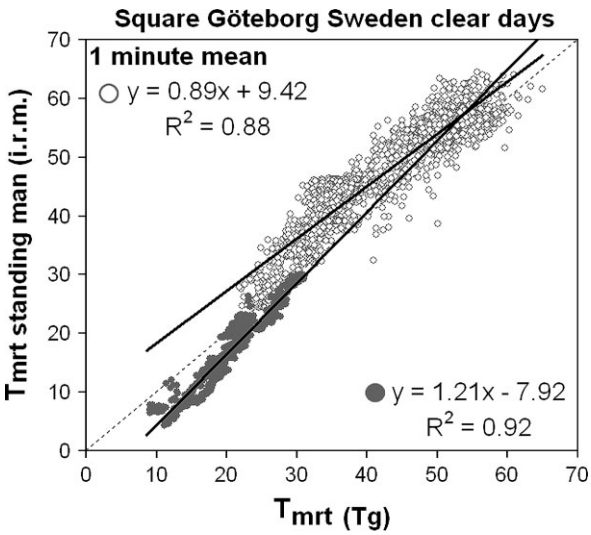

Figure 7. Mean radiant temperature of a standing man ($T_{mrt\ standing\ man\ (i.r.m.)}$) (Method A), *versus* $T_{mrt\ (Tg)}$ (Method B) during clear weather conditions. Filled circles represent shaded conditions and open circles represents non-shaded conditions.

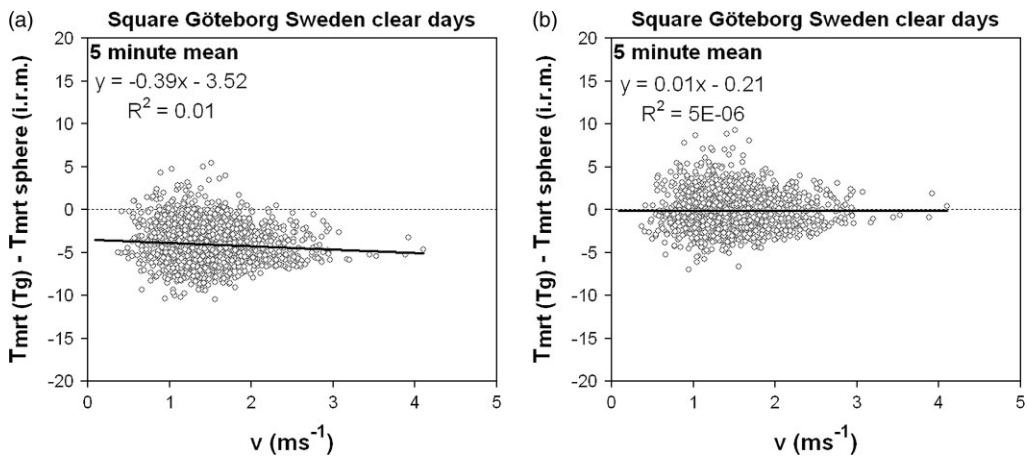

Figure 8. The influence of the 38 mm flat grey globes material characteristics. (a)The difference between $T_{mrt\ (Tg)}$ and $T_{mrt\ sphere\ (i.r.m.)}$ *versus* the wind speed according to Equation (3) (ASHRAE, 2001). (b)The difference between $T_{mrt\ (Tg)}$ and $T_{mrt\ sphere\ (i.r.m.)}$ *versus* the wind speed using corrected values of mean convection coefficient (Equation 4).

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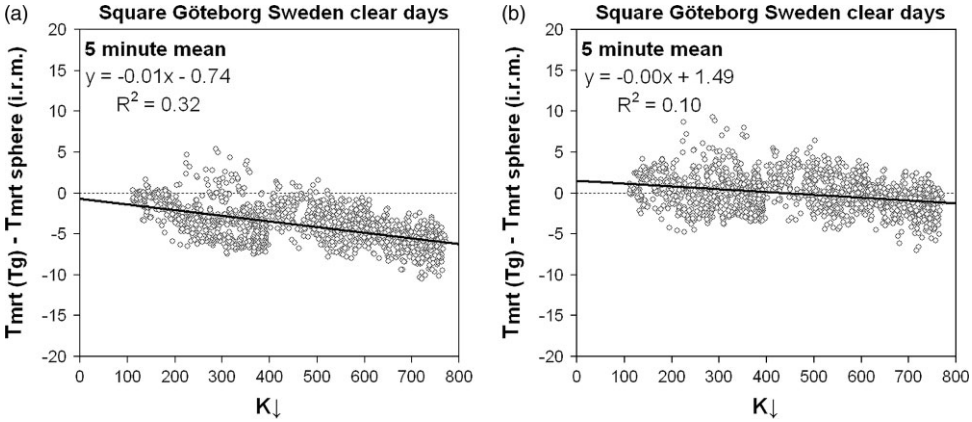

Figure 9. (a)The difference between $T_{\text{mrt (Tg)}}$ and $T_{\text{mrt sphere (imr)}}$ *versus* the $K\downarrow$ according to Equation (3) (ASHRAE, 2001). (b)The difference between $T_{\text{mrt (Tg)}}$ and $T_{\text{mrt sphere (imr)}}$ *versus* $K\downarrow$ using corrected values of mean convection coefficient (Equation 4).

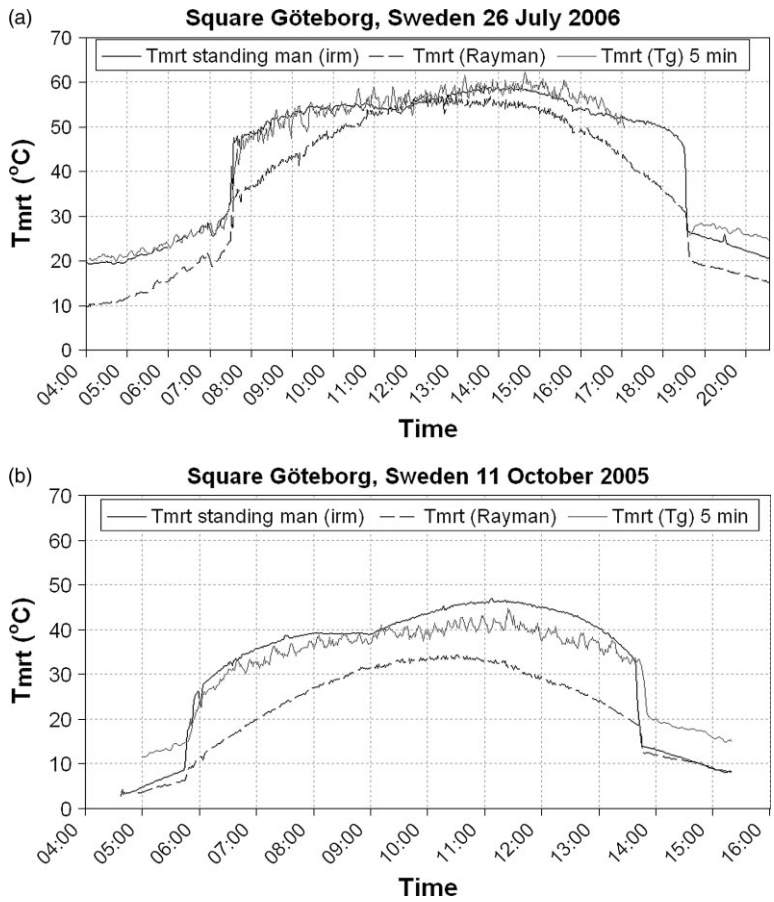

Figure 10. Mean radiant temperature determined by integral radiation measurements and angular factors, $T_{\text{mrt standing man (i.r.m.)}}$, 38 mm flat globe temperature measurements, $T_{\text{mrt (Tg)}}$ using corrected values of mean convection coefficient (Equation 4) and Rayman 1.2 software, $T_{\text{mrt (Rayman 1.2)}}$ (a) on 26 July 2006 and (b) on 11 October 2005.

DISCUSSION

The standard 150 mm copper globe thermometer takes up to 20 min to reach equilibrium (McIntyre, 1980). If the air speed or temperature changes over that time then equilibrium is never reached, which introduces an element of uncertainty in the T_{mrt} value. A test chamber study performed by Hey (1968) showed that equilibrium is reached more quickly if a smaller globe is used. However, by reducing the size of the sphere enclosing the thermometer bulb, the convective transfer coefficient increases and the proportional effect of radiation on the final temperature is reduced (McIntyre, 1980). A smaller globe diameter will thus affect the air temperature and air velocity, reducing the accuracy of the measurement of the T_{mrt} (Olesen *et al.*, 1989). A balance between response time and accuracy is required in order to identify an optimum size. The 38 mm flat grey globe thermometer used in this study has a response time of less than 5 min

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based on indoor tests (e.g. Nikolopoulou *et al.*, 1999). In outdoor settings, this study shows that the difference between Method A (integral radiation measurements) and Method B (38 mm flat grey globe thermometer) decreases substantially when using 5 min mean values (Figure 5). Using 10 min mean values only decreases the scattering slightly. Results from this study thus show that most of the effects of radiation, temperature and wind speed changes are smoothed out by using 5 min mean values. When the T_{mrt} departs from the ambient air temperature by only a few degrees, e.g. in shaded conditions (Figure 7), and the changes in the radiation fluxes, air temperature and air speed are small over time equilibrium is reached within a few minutes. In these circumstances, mean averages lower than 5 min can be used. However, when the T_{mrt} deviates from the ambient air temperature by several degrees and the radiation fluxes, air temperature and air speed change rapidly over time, 10 min average should be considered.

The T_{mrt} is defined with respect to the body under investigation. The shape of the sensor is thus a factor. As shown in Figure 6(a), a sphere shape theoretically underestimates the T_{mrt} of a standing person when the $K\updownarrow/K_{tot}$ is less than 0.36, i.e. the proportion of the vertical radiation is low and thus the angle of the incident solar radiation. The results presented in Figure 6(b) show that the difference between Method A (standing man) and Method B (38 mm flat grey globe thermometer) is 3.8 K, due to the difference in shape over the whole range of $K\updownarrow/K_{tot}$ (0.15–0.45). The results from this study show that although an ellipsoid–shaped sensor, which gives a closer approximation of the human shape (Olesen *et al.*, 1989), would probably give a more accurate estimation of the T_{mrt} of a standing man, the spherical shape of the globe thermometer works rather well. At high latitudes, the elevation of the sun is low during much of the year. This results in a fairly large amount of radiation from the four cardinal points (K_{east}, K_{west}, K_{south}, K_{north}) in comparison to the radiation from the upper hemisphere ($K\downarrow$). The radiation from the four cardinal points results in a relatively high T_{mrt} for a standing man (greater projected area to the side than to the sky and ground), even when the incoming solar radiation is low (Figure 4).

By systematically and empirically changing the mean convection coefficient resulting in Equation (4), the accuracy of Method B was improved. The corrected mean convection coefficient is more representative of the 38 mm flat grey globe thermometers material characteristics (heat storage and conductivity) and size than the original one. However, the influence of paint thickness demands an individual calibration of each globe thermometer to achieve higher accuracy. This study shows that the difference between Method A and B is reduced to less than ± 3.5 K when using Equation (4) and making allowances for the shape. These results are valid in conditions with air velocity between 0.1 and 4.0 ms^{-1} and incoming short-wave radiation ranging between 100 and 850 Wm^{-2}. The remaining error (Figure 9) can be assigned to instrumentation errors from the 38 mm flat grey globe thermometer and radiation instruments, and the response time and albedo of the globe. The flat grey colour of the globe thermometer is supposed to represent the radiant properties of the skin and general clothing of a person. As shown in Figure 7, the flat grey colour slightly overestimates the T_{mrt} during shady conditions and slightly underestimated it in non-shady conditions. Previous studies have shown that the standard black-coloured globe overestimates the influence of short-wave radiation and that a flat, grey coloured globe better represents the radiation characteristics of normal clothing (Olesen *et al.*, 1989). The results of this study show that a slightly lower albedo of the globe thermometer could further improve the results obtained from Method B.

The results presented in Figure 10 show that the difference between Method A (integral radiation measurements) and Method B (38 mm flat grey globe thermometer) using Equation (4) is generally relatively small during the whole day, particularly in the transition from shady to non-shady conditions. However, Method B gives a relatively large scatter in T_{mrt}. This is because the air temperature and wind speed are measured instantly, while there is a delay in the 38 mm flat grey globe thermometer response. Furthermore, the air and wind sensor is located close to, but not exactly at the same position as the globe thermometer. The difference between Methods A and B is smoothed out when Equation (4) is used. However, in October, Method B slightly underestimates T_{mrt}; this is probably due to differences related to low sun elevation and shape. The local minimum in T_{mrt} during the day using Method A (Figure 4(c) and (f)) is an artefact of the orthogonal instrument setup (i.e. increased mean instrumental error with high angles of incidence Kipp & Zonen, 2002). Thus, the daily T_{mrt} pattern is given more accurate using Method B compared to Method A.

The study shows that the Rayman model (Method C) works very well during the middle of the day in the summer, i.e. at high sun elevations (Figure 10(a)). However, the model underestimates T_{mrt} considerably in the morning and afternoon, i.e. at low sun elevations. As shown in Figure 10(b), the Rayman 1.2 also underestimates T_{mrt} at noon in October, which also can be interpreted as a result of the low sun elevation. Multiple reflections of short-wave radiation and the emittance of long-wave radiation from the surrounding surfaces are crucial to the estimation of T_{mrt} at low sun elevations. It is not clear how these aspects are incorporated into the Rayman model from related literature. For a high latitude city such as Göteborg, this means that Method C underestimates T_{mrt} during much of the year (autumn, winter and spring) as well as in the mornings and evenings in the summer.

The 38 mm flat grey globe thermometer is tested in an environment with a high sky view factor and extensive, homogenous surfaces. Studies by Thorsson *et al.* (2006) indicate that this type of globe thermometer also works well in more complex urban settings. The performance of the 38 mm flat grey globe thermometer in other environments however needs to be further studied. Another important thrust of further studies could be to

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analyse data from a wide range of wind, air temperature and radiation measurements.

This study has shown that the 38 mm flat grey globe thermometer is an easy and accurate method for estimating the T_{mrt} in an outdoor urban setting. Furthermore, this type of globe thermometer is a mobile and cheap instrument. In spite of this, it is seldom used in studies of outdoor comfort. One reason for this may perhaps be the absence of outdoor validation data. Urban planners and designers always ask for good and simple tools to estimate thermal comfort. Given the results above, the 38 mm flat grey globe thermometer can be employed for this purpose.

CONCLUSIONS

The objective of this study was to compare three different methods for estimating the T_{mrt} in the outdoor urban setting, including: (a) integral radiation measurements, (b) 38 mm flat grey globe thermometer and (c) Rayman 1.2 software.

The study shows that the difference between Method A and Method B was relatively small. Most of the discrepancy, which was due to rapid changes in radiation, temperature and wind speed, was smoothed out using 5 min mean values. The accuracy of Method B was improved by systematically and empirically changing the mean convection coefficient. The new coefficient is valid in conditions with air velocity ranging between 0.1 and 4.0 ms^{-1} and incoming short-wave radiation ranging between 100 and 850 Wm^{-2}. The study also shows that the flat grey colour of the globe slightly underestimates short-wave radiation (i.e. sunshine) and that Method B could be further improved through the use of a colour with a slightly lower albedo. Furthermore, the study shows that Method C works very well during the middle of the day in the summer, i.e. at high sun elevations. However, the model considerably underestimates the T_{mrt} in the morning and afternoon and in the autumn, i.e. at low sun elevations.

By applying the new mean convection coefficient, the 38 mm flat grey globe thermometer can successfully be used to estimate the T_{mrt} in the outdoor setting. Furthermore, the 38 mm flat grey globe thermometer is a simple, mobile and cheap instrument and is thus a valuable tool for thermal comfort researchers or urban planners and designers.

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REFERENCES

Ali-Toudert F, Mayer H. 2005. Outdoor thermal comfort in the old desert city of Beni-Isquen, Algeria. *Climate Research* **28**: 243–256.

ASHRAE. 2001. ASHRAE Fundamentals Handbook 2001 (SI Edition) American Society of Heating, Refrigerating, and Air-Conditioning Engineers, ISBN: 1883413885.

Azar S. 2006. TownScope III, University of Liege, Belgium. http://www.townscope.com.

Bedford T, Warner CG. 1934. The globe thermometer in studies of heating and ventilation. *Journal of Instrument Hygiene* **34**: 428–432.

Bruse M. 1999. Die Auswirkungen kleinskaliger Umweltgestaltung auf das Mikroklima-Entwicklung des prognostischen numerischen Modells ENVI-met zur Simulation der Wind-, Tempertaur-, und Feuchteverteilung in städtischen Struckturen, PhD thesis, University Bochum, Germany.

Bruse M. 2006. *ENVI-met 3 – a three dimensional microclimate model*. Ruhr Universität Bochum, Geographischer Institut, Geomatik. http://www.envi-met.com.

de Dear R. 1987. Ping-pong globe thermometers for mean radiant temperatures. *Heating and Ventilation Engineer and Journal of Air Conditioning* **60**: 10–11.

Fanger PO. 1972. Thermal comfort. *Analysis and Application in Environment Engineering*. McGraw Hill: New York.

Hey EN. 1968. Small globe thermometers. *Journal of Scientific Instruments (Journal of Physics Education)* Series 2, **1**: 955–957.

Höppe P. 1992. A new procedure to determine the mean radiant temperature outdoors. *Wetter unt Leben* **44**: 147–151.

Humphreys MA. 1977. The optimum diameter for a globe thermometer for use indoors. *Annals of Occupational Hygiene* **20**(2): 135–140.

Kipp & Zonen. 2002. *Instruction Manual for CNR1 Net-radiometer*. Kipp & Zonen. P.O. Box 507, Röntegenweg 1: Delft.

Kuehn LA, Stubbs RA, Weaver RS. 1970. Theory of the globe thermometer. *Journal of Applied Physiology* **29**: 750–757.

Matzarakis A. 2000. Estimation and calculation of the mean radiant temperature within urban structures. *Manual to RayMan*. University of Freiburg: Germany.

Matzarakis A. 2004. Personal communications. Meteorologisches Institut, Universitaet Freiburg, Werderring 10, D-79085 Freiburg, Germany.

Matzarakis A, Rutz F, Mayer H. 2000. Estimation and Calculation of the mean radiant temperature within urban structures. In *Biometeorology and Urban Climatology at the Turn of the Millennium*, Macquarie University, de Dear RJ, Kalma JD, Oke TR, Auliciems A (eds).: Sydney; 273–278, ICB-ICUC'99, WCASP-50, WMO/TD No 1026.

Mayer H, Höppe P. 1987. Thermal comfort of man in different urban environments. *Theoretical and Applied Climatology* **38**: 43–49.

McIntyre DA. 1980. *Indoor Climate*. Architectural Publication Series: London.

Nikolopoulou N, Baker N, Steemers K. 1999. Improvements to the globe thermometer for outdoor use. *Architectural Science Review* **42**: 27–34.

Nikolopoulou N, Baker N, Steemers K. 2001. Thermal comfort in outdoor urban spaces: understanding the human parameter. *Solar Energy* **70**: 227–235.

Olesen BW, Rosendahl J, Kalisperis LN, Summers LH, Steinman PE. 1989. Methods for measuring and evaluating the thermal radiation in a room. *ASHRAE Transactions* **95**: 1028–1044.

Spagnolo JC, de Dear RJ. 2003. A field study of thermal comfort in outdoor and semi-outdoor environments in subtropical Sydney Australia. *Building and Environment* **38**: 721–738.

Teller J, Azar S. 2001. TownScope II – a computer system to support solar access decision-making. *Solar Energy* **70**: 187–200.

Thorsson S, Lindberg F, Eliasson I, Holmer B. 2006. Measurements of mean radiant temperature in different urban structures. In *6th International Conference on Urban Climate*, Lindqvist S, Grimmond CBS (eds). Urban Climate Group, Department of Geosciences, Göteborg University: Sweden; 867–870.

VDI. 1994. Environmental meteorology, interactions between atmosphere and surface; calculation of short-and long wave radiation. Part I: Climate, VDI 3789, Part 2: VDI/DIN- Handbuch Reinhaltung der Luft, Band 1b, Düsseldorf.

Vernon HM. 1932. The measurement of radiant temperature in relation to human comfort. *Journal of Industrial Hygiene* **14**: 95–111.

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